

Computer Systems II

CSC 2405 - Spring 2025

Logistics

- Section 1 meeting time: MWF 8:30 – 9:20 AM in Mendel G87
- Section 2 meeting time: MWF 9:35 – 10:25 AM in Mendel G87

Contact Information

- Instructor: Venkat Margapuri
- Email: vmargapu@villanova.edu
- Office Location: 165B Mendel Hall
- Office Hours: Mon Wed 10:30 AM – 11:30 AM, and by appointment; in-person and Zoom: <https://villanova.zoom.us/j/8319414991>
- Teaching Assistant: Anmol Gautam
- Email: agauta01@villanova.edu
- Office Location: 292 Mendel Hall
- Office Hours: Tue Thu 3:00 PM – 4:00 PM, and by appointment; in-person on Tuesdays and Zoom: <https://villanova.zoom.us/j/8319414991> on Thursdays (available in-person on Thursdays upon student request)

Course Description

This course is the second part of a two-semester sequence, Computer Systems I and II. The goal of this course is to bridge the gap between high-level programming and actual computer components, such as processors, caches, operating systems, compilers, linkers, and so on. This course will teach the way computer systems work from the low-level device interrupts up to processes, threads, memory management, file systems and security. One main aspect of this course will be programming projects, which will enable hands-on learning while gaining valuable and practical experience in computer systems. Note that this is a class on systems concepts, not on C programming. Most of the basic constructs such as if statements, loops, function syntax, switch case statements and operators are the same in C and Java, and you are expected to know them from CSC 1052.

Learning Objectives

1. Understand the objectives and functions of modern operating systems
2. Describe and use techniques for achieving concurrency and synchronization
3. Explain the workings of a system with virtual memory management
4. Understand basics of networking and Internet communication
5. Describe the role of error correction codes in memories, storage and networks
6. Understand security risks and protection mechanisms for computer systems

Prerequisites

- CSC 2400 Computer Systems I

Course Resources

- All lecture material and recordings will be posted to Blackboard. Please visit regularly and make sure you are up to date.
- [Operating Systems: Three Easy Pieces](#), by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, online (free)

Course Requirements

- Homework Assigned weekly. Completing all assignments timely and diligently is crucial for mastering the subject.
- Tests: Midterm and one final presentation. No final exam. Tests will be closed books, closed notes, closed electronic devices. However, you are allowed to bring one sheet of paper (letter size) with any information you think will help you during the exam. Such notes may not be shared during the exam.
- Quizzes: Held every Monday on Blackboard during the first ten minutes of the lecture, unless stated otherwise.

Grading

Grade Distribution

Homework	40%
----------	-----

Midterm	25%
Final Presentation	25%
Quizzes	10%
Total	100%

Letter Grades

A	≥ 94 %
A-	≥ 90 %
B+	≥ 87 %
B	≥ 84 %
B-	≥ 80 %
C+	≥ 77 %
C	≥ 74 %
C-	≥ 70 %
D+	≥ 67 %
D	≥ 64 %

Course Policies

Class Attendance

Students are required to attend all the class meetings. If you cannot make a scheduled class for any reason, please contact the instructor ahead of time with proper documentation to supplement your claim.

In addition to the attendance policy stated above, students are entitled to two excused absences for any reason that may contribute to their personal wellness. Students must advise the instructor by email before class of their intent to utilize a Personal Day as the reason for their absence. A Personal

Day will not be approved retroactively. Students may, but are not required, to provide additional information regarding their absence. Additionally, Personal Days may not:

- be used on consecutive class days;
- be used in the same week;
- be used immediately preceding or following a University holiday or break period;
- be used on days when exams, presentations or other major assignments are scheduled.

A Personal Day does not grant an automatic extension for items due. Students remain responsible for all assignments, exams, presentations, etc. due on that date. It is in the instructor's discretion to determine whether any extension is appropriate given individual circumstances.

Grading Policy

Assignments are due at the time and date indicated on each assignment. 20% of the points will be deducted for each working day (M – F) that an assignment is late. Thus, after one week of delay, the assignments are worth 0 points.

Makeup Assignments

There will be no makeup assignments, quizzes, or exams. If you need to miss an exam due to an emergency, try to notify me at least a week in advance so we can make arrangements to make it up. Makeup exams will not be easier than regularly scheduled exams.

Academic Integrity

All students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. For the College's statement on Academic Integrity, you should consult the Enchiridion. You may view the university's Academic Integrity Policy and Code, as well as other useful information related to writing papers, at the Academic Integrity Gateway web site: <http://library.villanova.edu/Help/AcademicIntegrity>.

Office of Access and Disability Services (ADS) and Learning Support Services (LSS)

It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. Go to the Learning Support Services website (<http://learningsupportservices.villanova.edu>) for registration guidelines and instructions. For physical access or temporarily disabling conditions, please contact the Office of Disability Services at [\(610\) 519-4095](tel:6105194095) or email stephen.mcwilliams@villanova.edu. Registration is needed in order to receive accommodations.

Absences for Religious Holidays

Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University's obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with their professors as soon as possible, normally at least two weeks in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence. <https://www1.villanova.edu/villanova/provost/resources/student/policies/religiousholidays.html>

Collaboration Policy

Discussion of techniques and ideas covered in class is encouraged but the homework assignments must reflect original work. No discussion will be allowed during the exam.

Artificial Intelligence Use on Assignments

The use of A.I.-generated content is allowed in this course. Even if you have significantly edited AI-generated material, you must identify the AI tool used to assist in generating your work. You are required to provide the name of the tool, date used, and prompts used to generate the output. If questions arise about the authorship of the submitted work, you are responsible for authenticating your authorship. You should save evidence of your authorial process, such as drafts, notes, version histories, and complete transcripts of AI assistance.

Tentative Schedule

The course outline below is approximate and subject to change as the semester progresses. It is the responsibility of the student to learn and adjust to changes.

Dates	Topic	Readings
1/13	Goals and Introduction	OSTEP Chapter 2, Introduction
1/15	Basics of I/O and Multiprogramming Unix Process Management	OSTEP Chapter 36, I/O Devices
1/17	Basics of I/O and Multiprogramming Unix Process Management	OSTEP Chapter 36, I/O Devices
1/22	Creating and Destroying Unix Processes	OSTEP Chapter 4, The Abstraction: The Process
1/24	Creating and Destroying Unix Processes	OSTEP Chapter 4, The Abstraction: The Process
1/27	Execution Environment	OSTEP Chapter 5, Process API (up to section 5.3)
1/29	Executing Processes	
1/31	Hands-On: exec system calls	
2/3	Inter-Process Communication	
2/5	Inter-Process Communication	
2/7	Inter-Process Communication	
2/10	Scheduling	OSTEP Chapter 7, Scheduling

Dates	Topic	Readings
2/12	Scheduling	OSTEP Chapter 18, Paging
2/14	Scheduling	OSTEP Chapter 18, Paging
2/17	Memory Management	OSTEP Chapter 19, Translation Lookaside Buffer
2/19	Memory Management	OSTEP Chapter 19, Translation Lookaside Buffer
2/21	Virtual Memory	OSTEP Chapter 22, Page Replacement Policies
2/24	Virtual Memory	OSTEP Chapter 22, Page Replacement Policies
2/26	Midterm Exam Review	
2/28	Midterm Exam	
3/3	Spring Break	
3/5	Spring Break	
3/7	Spring Break	
3/10	Threads	OSTEP Chapter 26, Concurrency and Threads
3/12	Threads	OSTEP Chapter 27, Thread API
3/14	Hands-on	
3/17	Semaphores	OSTEP Chapter 31, Semaphores
3/19	Semaphores	
3/21	Hands-on	
3/24	Classic Synchronization Problems	

Dates	Topic	Readings
3/26	Classic Synchronization Problems	
3/28	Hands-on	
3/31	IPC with Sockets	
4/2	IPC with Sockets	
4/4	Hands-on	
4/7	File Systems	
4/9	File Systems	
4/11	Hands-on	
4/14	RAID	
4/16	Course review and catchup	
4/23 – 4/30	Final Presentations	